diff-diff

Difference-in-differences when units are treated in different years

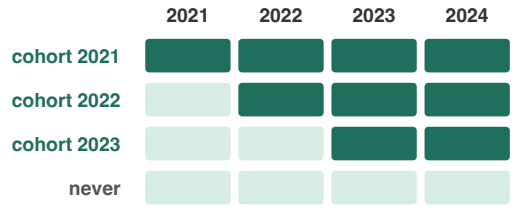
Python cheatsheet · MATH60033A · Quantitative Methods in IB · used in session 11

```
import diff_diff as dd
```

pip install diff-diff · the Python counterpart of R's did and HonestDiD

1 · Shape the panel

Long format: one row per unit and year, plus one column saying when each unit was first treated.



dark = treated · light = untreated, usable as a control

cohort 2021 has no untreated year: it cannot be used

`adopted = d[d["treated"] == 1].groupby("geo")["time"].min()`

the adoption year, from the **full** treatment series — not from the rows that happen to have an outcome.

`d["g"] = d["geo"].map(adopted).fillna(0).astype(int)`

`first_treat`: the adoption year, and **0** for units never treated.

> Count the cohorts and their pre-periods **before** fitting: a cohort with no untreated year is no evidence, and a cohort of one is one unit's story.

2 · Fit Callaway–Sant'Anna

`cs = dd.CallawaySantAnna(cluster="geo")`

each cohort against **clean** controls (never or not-yet treated), standard errors clustered by unit.

`cs = cs.fit(d, outcome="y", unit="geo", time="time", first_treat="g")`

the column names, said out loud.

`cs.overall_att, cs.overall_se, cs.n_clusters`

the average effect on the treated, its standard error, the number of clusters.

> Report it **beside** two-way fixed effects (`smf.ols("y ~ D + C(geo) + C(time)")`). If they agree, show it; if not, that is the finding.

3 · Read the group-time effects

`gt = cs.to_dataframe()`

one row per (cohort, year): `group`, `time`, `effect`, `se`.

`gt["skip_reason"]`

why a cell was **dropped**: `missing_period` (no pre-period), `zero_treated_control` (nobody to compare).

`gt[gt["time"] < gt["group"]]`

the rows **before** treatment: pre-trend evidence, not effects.

> A one-unit cohort can show a *smaller* standard error than a large one. That is not precision: there is no variation left to measure.

4 · Aggregate

`cs.aggregate("group").to_dataframe()`

one effect per cohort.

`es = cs.aggregate("event_study")`

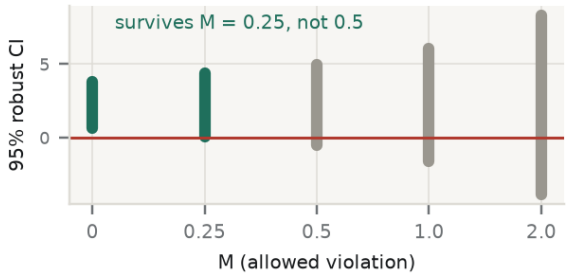
effects by years since adoption: the event study.

`es.to_dataframe().query("event_time < 0")`

the pre-treatment points; the reference year is zero by construction.

5 · How much violation does it survive?

Rambachan and Roth (2023): do not test parallel trends; ask how much of it the result needs.



`h = dd.HonestDiD(method="relative_magnitude", M=0.5).fit(es)`

allow post-treatment departures up to M times the largest pre-treatment one.

`h.ci_lb, h.ci_ub, h.is_significant`

the robust 95% interval, and whether it still excludes zero.

> Report the **breakdown** value: the M at which the interval first reaches zero.

6 · When diff-diff complains

`KeyError on a column name`

`outcome`, `unit`, `time`, `first_treat` must name columns of the data.

`a cohort is missing from the output`

it had no pre-period or no control: read `skip_reason`, and say so in the report.

`the never-treated coded as 9999`

use **0**: any other number is read as an adoption year.